

Ashley N. Prow-Fleischer

last update: Aug. 26

Email: prowa@stanford.edu

ORCID: <https://orcid.org/0000-0003-3371-2453>

Personal Website: aprow620.github.io

Education & Training

- 2025 - Now Postdoctoral Scholar Stanford University | [Paleobiology Group](#) | PI. Dr: Jonathan Payne
- 2021 - 2025 Ph.D. Syracuse University | Paleoclimatology | PI: Dr. Zunli Lu
- 2014 - 2017 B.S. Florida State University | Environmental Chemistry
- 2014 - 2017 B.S. Florida State University | Biomathematics

Publications

Peer-Review Journals

Prow-Fleischer, A., Lu Z., Penn, J.L., Deutsch, C., Todes, J.P., Blättler, C.L., Day, J., Mucciarone D., Pohl, A., Ridgwell, A., Al Aswad J. A., Guthrie, R., Jenkyns, H.C., Robinson, S.A., Yanan Shen, Y., Rationale, D., He, R., Elrick, M., Payne, J. L. (**in submission**). Temperature-dependent physiological traits and biogeography link Late Devonian Kellwasser Extinction to climate cooling.

Bazzi, M., Prow-Fleischer, A., Siverson, M., Al Aswad, J.A., Ebersole, J.A., Payne, J.L. (**in review**). Broad-scale biogeographic stability despite mass extinction in sharks across the K/Pg boundary. *Proc. R. Soc. B*

He, R., Pohl, A., Zhang, X., Chang, C., Prow-Fleischer, A. Payne, J.L., Xiao, S., Ridgwell, A., Lu, Z. (2026) A reversed latitudinal ocean oxygen gradient in the Proterozoic Eon. *Nat. Geosci.* **19**, 325–330. <https://doi.org/10.1038/s41561-025-01896-w>

Prow-Fleischer, A.N., & Lu, Z. (2025). A laser ablation ICP-MS protocol for high-resolution I/Ca profiling in corals. *Rapid Comm. Mass Spec.* <https://doi.org/10.1002/rcm.10002>

Prow-Fleischer, A.N., Lu, Z., Blättler, C.L., He, T., Yang, Z., Singh, P., Kemeny, P.C., Todes, J.P., Pohl, A., Bhattacharya, T., van de Schootbrugge, B., Wignall, P.B., Payne, J.L. (2025). Calcium isotopes support spatial redox gradients on the Tethys European margin across the Triassic-Jurassic boundary. *Chemical Geology*. <https://doi.org/10.1016/j.chemgeo.2024.122530>

Prow-Fleischer, A.N., Lu, Z., Meehan, K.C., Yang, Z., Ivany, L.C., Payne, J.L., (2024) Body-size reductions in dacroconarid tentaculitoids during Late Devonian warming. *Geosphere*. <https://doi.org/10.1130/GES02759.1>

- He, R., Pohl, A., **Prow, A.N.**, Jiang, G., Huan C. C., Saltzman, M., Ridgwell, A., Lu, Z. (2024). The dynamic ocean redox evolution during the Late Cambrian SPICE event: evidence from the I/Ca proxy. *Global and Planetary Change*. <https://doi.org/10.1016/j.gloplacha.2024.104354>
- Lu, Z., Rickaby, R.E.M., Payne, J.L., **Prow, A.N.** (2024). Phanerozoic Coevolution of O₂-CO₂ and Ocean Habitability. *National Science Review*. <https://doi.org/10.1093/nsr/nwae099>
- Prow, A.N.**, Lu, Z., Frappier, A., Weisbeck, L., Underwood, C. (2023). Extraction of calcareous Dacryoconarid microfossils from limestones and mudrocks by surfactants paired with freeze-thaw processing. *Marine Micropaleontology*. <https://doi.org/10.1016/j.marmicro.2023.102216>
- Lu, Z., Thomas, E., Rickaby, R.E.M., Lu, W., and **Prow, A.N.** (2023). Commentary: Planktic foraminifera iodine/calcium ratios from plankton tows: *Frontiers in Marine Science*, <https://doi.org/10.3389/fmars.2023.1221835>

Grants and Fellowships (* competitive | ** collaborative)

- | | |
|-------------|--|
| 2026 | National Science Foundation ACCESS Explorer's Grant (EES260080 & CIS260896) 100 core hours** |
| 2025 | The Micropaleontology Society Angelina Messina Grant \$600 |
| 2024 - 2025 | Research Excellence Doctoral Funding Graduate Fellowship (Syracuse University)* |
| 2024 | Geological Society of America (GSA) On-to-the-Future Mentor Program Travel \$600 |
| 2024 | GSA/Chevron Field Trip Grant \$290 |
| 2024 | William B. & Dorothy Heroy Research Grant \$2200 |
| 2023 | Syracuse University Graduate Student Organization Travel Grant \$600 |
| 2023 | Travel Grant Pre-GSA workshop, EarthCube Geochronology \$800 |
| 2023 | Summer Research Grant, Merriam Endowment, Syracuse University Internal \$2000 |

Presentations

Invited Talks

- | | |
|------------|---------------------------------------|
| April 2027 | Environmental Fridays Season 12 TBD |
|------------|---------------------------------------|

Feb 2026 Rochester Academy of Science Paleontology Division | “Linking anoxia, climate cooling, and marine extinction selectivity across the Kellwasser events”

Conference Talks

July 2025 Prow-Fleischer, A.N., Lu, Z., Blättler, C., Ridgwell, A., Payne, J., Penn, J., Al Aswad, J., Pohl, A. “Linking anoxia, climate cooling, and marine extinction selectivity across the Kellwasser events using proxy-constrained Earth-system modelling approach”.
Goldschmidt 2025, Prague, Czech Republic

April 2024 Prow, A.N., Lu, Z., Blättler, C.L., He, T., Yang, Z., Singh, P., Kemeny, P.C., Todes, J.P., Pohl, A., Bhattacharya, T., van de Schootbrugge, B., Wignall, P.B., Todaro, S. Payne, J.L. “Calcium isotopes validate spatial redox gradients on the Tethys European margin across the Triassic-Jurassic boundary.” NE Geobiology Symposium, New Haven, CT.

Dec.2023 Prow, A.N., Lu, Z., Blättler, C.L., He, T., Yang, Z., Singh, P., Kemeny, P.C., Todes, J.P., Pohl, A., Bhattacharya, T., van de Schootbrugge, B., Wignall, P.B., Payne, J.L.
“Temporal and spatial dynamics of paleo-redox conditions across the Triassic-Jurassic boundary.” American Geophysical Union Annual Meeting. San Francisco, CA

Oct. 2023 Prow, A.N., Lu, Z., Meehan, K.C., Yang, Z., Ivany, L.C., Payne, J.L., “Dacryoconarid Body Size Trends across the Late Devonian Punctata Event: Insights into their Evolutionary Responses to Paleoenvironmental Shifts.” Geological Society of America Connects , Pittsburg, PA

Aug. 2023 Prow, A.N., Lu, Z., Meehan, K.C., Yang, Z., Ivany, L.C., Payne, J.L., “Exploring Dacryoconarid Body Size Trends across the Late Devonian Punctata Event: Insights into their Evolutionary Responses to Paleoenvironmental Shifts.” Paleontological Research Institute Summer Symposium

Posters

Sept. 2024 Prow-Fleischer, A.N. & Lu, Z. “A laser ablation ICP-MS protocol for high-resolution I/Ca profiling in corals.” GSA Connects Annual Conference, Anaheim, CA

Sept. 2023 Prow, A.N., Lu, Z., Meehan, K.C., Yang, Z., Ivany, L.C., Payne, J.L., “Dacryoconarid body size and diversity trends across the *punctata* carbon isotope excursion, Appalachian Basin” International Sub-Commission for Devonian Stratigraphy conference, Genesee NY.

Student Presentations

- Dec. 2026 Mu, J.K., Prow-Fleischer, A., Payne, J.L. “Applying Distance Based Sensitivity Analysis to Quantify Variable Uncertainty in a Carbon-Calcium Isotope Mass Balance Model”
AGU Annual Meeting, San Fransico, Ca
- Dec. 2026 Kwak, Z., Prow-Fleischer, A., Payne, J.L. “Phosphorus Cycle is an Important Component in Reconciling Ca-C Isotope Records Across Late Devonian Upper Kellwasser Event”
AGU Annual Meeting, San Fransico, Ca

Mentoring

Research Mentor

Summer 2026 Stanford Young Investigators

Teaching Experience

Instructor

Spring 2026 Topics in Geobiology | Stanford University

Summer 2024 Instructor: Climate Change: Past, Present and Future | Syracuse University

Graduate Teaching Assistant

Spring 2022 Climate Change Past and Present

Fall 2021 Introduction to Oceanography,

Professional Teaching

2018 - 2021 Florida State University Department of Chemistry

Courses instructed: Introduction to Analytical Chemistry Laboratory, Advanced (Instrumental) Analytical Chemistry Laboratory, Physical Chemistry Laboratory

Professional Experience

2018 – 2021 Undergraduate Chemistry Laboratory Teaching Specialist | Florida State University

2017 - 2018 Lead Lab Technician | Surterra Wellness

Professional Service

Professional Societies

- April 2025 - Geological Society of America Student Advisor Council member (2 year term)
- Sept 2024 Organizer of Geological Society America '24 (Anaheim) Topical Sessions Part I and II
“Piecing together the extinction puzzle: Integrated approaches to quantifying organism-environment interactions.”

Ad-hoc Reviewer

Marine Micropaleontology, Science Advances

Departmental Committees

- 2024 - 2025 Syracuse University Chief Diversity Officer, Department Graduate Student Organization (GeoGo)
- 2022 - 2023 Student Affairs Coordinator, Department Graduate Student Organization (GeoGo)
- 2022 - Member Diversity, Equity, and Inclusion Committee, Syracuse University Department of Earth and Environmental Sciences

University Organizations

- 2022 - 2024 Member of Women in Science and Engineering Future Professionals Program (WISE-FFP) at Syracuse University (Certificate conferred 2024)

Awards and Honors

- 2025 Syracuse University College of Arts and Sciences Doctoral Prize
- 2024 GSA Foundation Penrose Circle Heroy Student Award
- 2024 Marjorie Hooker Outstanding Thesis proposal
- 2024 Honorable Mention Syracuse University Dean’s Award
- 2023/24 Publication Award, Syracuse University Department of Earth & Environmental Sciences
- 2013 Women’s History Month Excellence in STEM award, Tallahassee Community College

Community Engagement

- 2022-2025 Board Member for Friends of Clark Reservation State Park

Certifications

2026 Certificate in High Performance Research Computing | Issued by Stanford University